University Core Facility Technology Landscape

Comparative Map: UNR vs UBC vs MIT vs Caltech — All Science & Engineering Core Facilities

Compiled April 2026

Technology Category	Sub-Category	UNR SIL (Your Access)	UNR Other Core Facilities	UBC	MIT	Caltech	Notes
VIBRATIONAL SPECTROSCOPY							
Vibrational Spectroscopy	FT-IR (Mid-IR, Near-IR, Far-IR)	✓	—	✓	✓	✓	SIL: Nicolet 6700, Nexus 470, Nicolet 380, Bruker Tensor 27
Vibrational Spectroscopy	IR Microscopy (Micro-FTIR)	✓	—	—	✓	—	SIL: Bruker Hyperion coupled to Tensor 27
Vibrational Spectroscopy	Raman Microscopy (Confocal)	✓	—	—	✓	✓	SIL: Renishaw inVia
Vibrational Spectroscopy	ATR Spectroscopy	✓	—	✓	✓	✓	SIL: Diamond ATR on multiple FT-IRs
ELECTRONIC SPECTROSCOPY							
Electronic Spectroscopy	UV-Vis Spectrophotometry	✓	—	✓	✓	✓	SIL: Lambda 750, UV-2550
Electronic Spectroscopy	UV-Vis-NIR (extended range)	✓	—	✓	✓	✓	SIL: Lambda 750 covers 190-3300 nm
Electronic Spectroscopy	Fluorescence Spectroscopy	✓	—	—	✓	✓	SIL: FluoroMax-3
Electronic Spectroscopy	Circular Dichroism (CD)	✓	—	—	✓	✓	SIL: Jasco J-1500
Electronic Spectroscopy	Polarimetry	✓	—	✓	—	—	SIL: Jasco P-2000
MASS SPECTROMETRY							
Mass Spectrometry	LC-MS (Quadrupole)	✓	✓	✓	✓	✓	SIL: Waters ZQ ESCi; also Env. Eng. Lab
Mass Spectrometry	LC-TOF-MS (High-Res Accurate Mass)	✓	—	—	✓	✓	SIL: Agilent 6230 TOF
Mass Spectrometry	MALDI-TOF	✓	—	—	✓	✓	SIL: Shimadzu MALDI-8020
Mass Spectrometry	GC-MS (Quadrupole/Ion Trap)	✓	✓	✓	✓	✓	SIL: Agilent 7890A/5975C; Env. Eng. has Waters GC-MS
Mass Spectrometry	LC-MS/MS (Triple Quad/Orbitrap)	—	✓	✓	✓	✓	UNR Proteomics: Orbitrap Eclipse, timsTOF Pro2
Mass Spectrometry	ICP-MS	—	—	—	✓	✓	Caltech WEL has 3 ICP-MS units; MIT.nano
Mass Spectrometry	SIMS / NanoSIMS	—	—	—	—	✓	Caltech: Cameca NanoSIMS 50L (50 nm resolution)
Mass Spectrometry	Isotope Ratio MS (IRMS)	—	✓	—	—	✓	UNR: Nevada Stable Isotope Lab; Caltech WEL
CHROMATOGRAPHY / SEPARATION							
Chromatography	HPLC (Analytical/Prep)	✓	✓	—	✓	✓	SIL: Agilent 1200 Series (dual stacks)
Chromatography	GC-FID	✓	—	—	✓	✓	SIL: Agilent 7890A standalone + chiral GC
Chromatography	Ion Chromatography (IC)	—	—	—	✓	✓	Caltech WEL: IC and IC-MS
NUCLEAR MAGNETIC RESONANCE (NMR)							
NMR Spectroscopy	Liquid-State NMR (≤400 MHz)	✓	—	✓	✓	✓	SIL: Bruker 400 MHz; UBC: Bruker 300 MHz; Caltech: 300/400 MHz
NMR Spectroscopy	Liquid-State NMR (500 MHz)	✓	—	—	✓	✓	SIL: Bruker 500 MHz; MIT DCIF; Caltech
NMR Spectroscopy	Liquid-State NMR (600+ MHz)	✓	—	—	✓	✓	SIL: Bruker 700 MHz w/ cryoprobe; MIT DCIF; Caltech 600 MHz
NMR Spectroscopy	Solid-State NMR (CP/MAS)	✓	—	—	✓	✓	SIL: Tecmag 400 MHz; MIT FBML; Caltech: 200/300/500 MHz
NMR Spectroscopy	Dynamic Nuclear Polarization (DNP)	—	—	—	✓	—	MIT Francis Bitter Magnet Lab
ELECTRON PARAMAGNETIC RESONANCE (EPR)							
EPR	EPR / ESR Spectroscopy	✓	—	—	✓	—	SIL: Bruker EMXPlus; MIT DCIF: Bruker EMX-Plus X-band
EPR	Variable-Temperature EPR (Cryogenic)	✓	—	—	✓	—	SIL: He recycling cryostat; MIT: 4.5K cooling
THERMAL ANALYSIS							
Thermal Analysis	DSC (Differential Scanning Calorimetry)	✓	—	—	✓	—	SIL: TA Instruments DSC Q20
Thermal Analysis	TGA (Thermogravimetric Analysis)	✓	—	✓	✓	—	SIL: TA Instruments TGA Q50; UBC: Netzsch TGA209
Thermal Analysis	Melting Point Determination	✓	—	—	—	—	SIL: OptiMelt
Thermal Analysis	Dynamic Mechanical Analysis (DMA)	—	—	—	✓	—	
Thermal Analysis	Rheometry (DSR)	—	✓	—	✓	—	UNR Western Regional Superpave Center
X-RAY DIFFRACTION & ANALYSIS							
X-Ray Techniques	Powder XRD	✓	—	✓	✓	✓	SIL: Bruker D8 Advance; Caltech: Rigaku SmartLab, PANalytical
X-Ray Techniques	Single-Crystal XRD	✓	—	✓	✓	✓	SIL: Bruker D8 Venture; MIT: Bruker APEX2; Caltech: Bruker D8
X-Ray Techniques	XRF (X-Ray Fluorescence)	—	—	—	—	✓	Caltech GPS: PANalytical Zetium
X-Ray Techniques	Synchrotron Beamline Access	—	—	—	—	✓	Caltech: dedicated SSRL beamline at Stanford
X-Ray Techniques	Macromolecular Crystallography	—	—	✓	✓	✓	Caltech Molecular Observatory; UBC CBR; MIT
ELECTRON MICROSCOPY							
Electron Microscopy	SEM (Scanning Electron Microscopy)	—	✓	✓	✓	✓	UNR EMMF: FEI Talos, Scios 2 FIB/SEM; Mackay: JEOL SEM
Electron Microscopy	TEM (Transmission Electron Microscopy)	—	✓	✓	✓	✓	UNR MCNV: JEOL JEM 2100F; UBC: multiple TEMs
Electron Microscopy	Cryo-EM (Structural Biology)	—	—	✓	✓	✓	UBC HRMEM: Titan Krios; MIT.nano; Caltech Beckman
Electron Microscopy	FIB-SEM (Focused Ion Beam)	—	✓	✓	✓	✓	UNR EMMF: Scios 2; UBC: Cryo-FIB-SEM
Electron Microscopy	EPMA (Electron Probe Microanalysis)	—	✓	—	—	✓	UNR Mackay: JEOL JXA iHP200F; Caltech GPS
Electron Microscopy	EDS/WDS (Elemental Analysis)	—	✓	✓	✓	✓	UNR EMMF/Mackay; Caltech GPS
SURFACE ANALYSIS & MICROSCOPY							
Surface Analysis	AFM (Atomic Force Microscopy)	✓	—	✓	✓	✓	SIL: Nanosurf EasyScan 2; UBC SIF; MIT.nano
Surface Analysis	STM (Scanning Tunneling Microscopy)	✓	—	✓	✓	—	SIL: EasyScan 2 AFM/STM; UBC Physics UHV-STM
Surface Analysis	XPS / ESCA	—	—	—	✓	—	MIT.nano surface analysis

Surface Analysis	Profilometry	—	—	—	✓	✓	MIT.nano; Caltech KNI
OPTICAL & FLUORESCENCE MICROSCOPY							
Optical Microscopy	Confocal Laser Scanning Microscopy	—	✓	✓	✓	✓	UNR CMI: Leica SP8; UBC LSI; Caltech BIF: 5 Zeiss LSMs
Optical Microscopy	Two-Photon / Multiphoton Microscopy	—	✓	✓	✓	✓	UNR CMI: Scientifica; Caltech: Leica STELLARIS 8 DIVE
Optical Microscopy	Light Sheet Microscopy (SPIM)	—	—	—	✓	✓	Caltech BIF: 2 LaVision/Zeiss systems
Optical Microscopy	Super-Resolution (STED, PALM, SIM)	—	—	✓	✓	✓	UBC LSI Imaging; MIT; Caltech: Airyscan 2
Optical Microscopy	Spinning Disk Confocal	—	—	✓	✓	✓	UBC BIF/LSI; Caltech BIF: Nikon
Optical Microscopy	FLIM (Fluorescence Lifetime)	—	—	—	—	✓	Caltech BIF: STELLARIS 8 FALCON
GENOMICS & SEQUENCING							
Genomics	Next-Gen Sequencing (Illumina Short-Read)	—	✓	✓	✓	✓	UNR Genomics Center: NextSeq 2000; Broad: NovaSeq
Genomics	Long-Read Sequencing (PacBio/Nanopore)	—	—	—	✓	—	MIT BioMicro Center
Genomics	Sanger Sequencing	—	✓	✓	—	—	UNR Genomics Center; UBC SBC
Genomics	Single-Cell Genomics / scRNA-seq	—	—	—	✓	✓	MIT BioMicro; Caltech SPEC
Genomics	qPCR / Fragment Analysis	—	✓	✓	✓	—	UNR Genomics: QuantStudio 3
PROTEOMICS & METABOLOMICS							
Proteomics	Quantitative Proteomics (LC-MS/MS)	—	✓	✓	✓	✓	UNR Proteomics: Orbitrap Eclipse, timsTOF Pro2
Proteomics	Metabolomics / Small Molecule Profiling	—	—	—	✓	—	MIT Whitehead Metabolite Profiling Core
Proteomics	Protein Crystallography	—	—	✓	✓	✓	Caltech Molecular Observatory; UBC CBR
FLOW CYTOMETRY & CELL ANALYSIS							
Cell Analysis	Flow Cytometry (Analysis)	—	✓	✓	✓	✓	UNR CACF; UBC ubcFLOW; Caltech: Beckman/Miltenyi
Cell Analysis	High-Speed Cell Sorting (FACS)	—	✓	✓	✓	✓	UNR CACF: BD sorters; UBC: 6 sorters; Caltech: 3 sorters
Cell Analysis	Mass Cytometry (CyTOF)	—	—	✓	—	—	UBC ubcCyTOF
Cell Analysis	Seahorse Metabolic Analysis	—	✓	—	—	—	UNR Cell Metabolism Core: XF24, XFp
NANOFABRICATION & CLEANROOM							
Nanofabrication	Photolithography	—	✓	✓	✓	✓	UNR Davidson Cleanroom; UBC AMPEL; MIT.nano/MTL
Nanofabrication	E-Beam Lithography	—	—	✓	✓	✓	UBC AMPEL/QMI; MIT.nano; Caltech KNI
Nanofabrication	Thin Film Deposition (PVD/CVD)	✓	✓	✓	✓	✓	SIL: Denton evaporator; UNR Cleanroom; MIT.nano
Nanofabrication	Etching (Wet/Dry/RIE)	—	✓	✓	✓	✓	UNR Davidson Cleanroom; MIT.nano; Caltech KNI
Nanofabrication	MBE (Molecular Beam Epitaxy)	—	—	✓	✓	—	UBC Physics; MIT
Nanofabrication	Microfluidics / Soft Lithography	—	—	✓	✓	✓	UBC AMPEL; Caltech KNI foundry
SYNTHESIS & REACTION SUPPORT							
Synthesis	Microwave Synthesis	✓	—	—	—	—	SIL: CEM Discover
Synthesis	Ozone Generation (Ozonolysis)	✓	—	—	—	—	SIL: Orec Ozonator
Synthesis	High-Pressure Apparatus	—	—	—	—	✓	Caltech GPS: piston-cylinder, multi-anvil, diamond anvil
Synthesis	Fume Hood / Wet Chemistry	✓	—	✓	✓	✓	SIL: Full fume hood station
MATERIALS TESTING & FABRICATION							
Materials	Spot Welding / Micro-Joining	✓	—	—	—	—	SIL: Hughes HRW 250B
Materials	KBr Pellet Press	✓	—	—	—	—	SIL: Hydraulic Press
Materials	Four-Point Probe (Sheet Resistance)	✓	—	—	✓	—	SIL: Four-Point Probe; MIT.nano
Materials	Mechanical Testing (Tensile/Hardness)	—	✓	✓	✓	✓	UNR ME Lab; MIT DMSE
Materials	CNC Machining / Machine Shop	—	✓	✓	✓	—	UNR ME Manufacturing Lab; UBC MAL; MIT
Materials	Friction Stir Welding	—	✓	—	—	—	UNR ME Lab
NUCLEAR & RADIATION RESEARCH							
Nuclear	Research Nuclear Reactor	—	—	—	✓	—	MIT MITR: 6 MW reactor
Nuclear	Neutron Irradiation / Activation	—	—	—	✓	—	MIT MITR beam ports
Nuclear	Pulsed Power / Z-Pinch	—	✓	—	✓	—	UNR ZPPL: 2 TW Z-pinch; MIT PSFC
Nuclear	High-Power Laser Systems	—	✓	—	✓	—	UNR ZPPL: 50/100 TW lasers
BRAIN & MEDICAL IMAGING							
Brain Imaging	Human MRI (3T)	—	✓	—	✓	—	UNR Neuroimaging: 2x 3T (Philips, Siemens); MIT Martinos
Brain Imaging	Animal MRI (High-Field)	—	—	—	✓	—	MIT Martinos: 9.4T Bruker
Brain Imaging	EEG / MEG	—	✓	—	✓	—	UNR Neuroimaging; MIT Martinos
Brain Imaging	Near-Infrared Spectroscopy (fNIRS)	—	✓	—	✓	—	UNR; MIT Martinos
ASTRONOMY & OBSERVATORIES							
Astronomy	Optical Telescope (Campus)	—	✓	—	—	—	UNR MacLean Observatory: 22" & 14" telescopes
Astronomy	Gravitational Wave Detection	—	—	—	✓	✓	LIGO: jointly operated by MIT & Caltech
Astronomy	Radio/Survey Telescopes	—	—	—	✓	✓	MIT: CHIME, Magellan, SPT, TESS affiliations
ENVIRONMENTAL & GEOSCIENCE							
Environmental	Stable Isotope Analysis	—	✓	—	—	✓	UNR Nevada Stable Isotope Lab; Caltech WEL
Environmental	Atmospheric Reaction Chambers	—	—	—	—	✓	Caltech Linde Center: 1,000 ft^3 chambers
Environmental	Water Quality / TOC-TIC Analysis	—	—	—	—	✓	Caltech WEL
Environmental	Seismic / Shake Table Testing	—	✓	—	—	—	UNR CCEER: 5 shake tables incl. 6-DOF
SAMPLE PREPARATION & INFRASTRUCTURE							

Sample Prep	Ultrapure Water System	✓	—	✓	✓	✓	SIL: Smart2Pure Pro (18.2 MΩ·cm)
Sample Prep	N_2 Generation / Gas Supply	✓	—	✓	✓	✓	SIL: Parker N_2 generator + He cylinders
Sample Prep	Analytical Balances	✓	—	✓	✓	✓	SIL: Mettler AE 100 + Mettler Toledo
Sample Prep	UPS / Power Protection	✓	—	✓	✓	✓	SIL: APC Back-UPS 1500
BIOINFORMATICS & COMPUTING							
Computing	Bioinformatics Analysis Platform	—	✓	✓	✓	—	UNR Nevada Bioinformatics Center; UBC SBC
Computing	High-Performance Computing (HPC)	—	✓	✓	✓	✓	UNR ZPPL cluster; MIT Lincoln Lab Supercomputing
BIOLOGICAL MODEL SYSTEMS							
Biology	Transgenic Mouse Facility	—	—	—	✓	✓	MIT Koch; Caltech
Biology	Zebrafish / Aquatic Models	—	—	✓	—	✓	UBC InSEAS/AERL; Caltech Beckman zebrafish
Biology	BSL-3 Containment Lab	—	✓	—	✓	—	UNR School of Medicine BSL-3
Biology	Greenhouse / Field Research	—	✓	—	—	—	UNR Valley Road: 29,280 sq ft greenhouse

University of Nevada, Reno — All Core Facilities

Facility Name	Department / College	Key Technology Categories	Major Instruments	Location	Access Notes
CHEMISTRY (SIL — Your Current Access)					
Chemistry SIL (Room 016)	Chemistry	FT-IR, Raman, UV-Vis, Fluorescence, CD, Polarimetry, LC-MS, TOF-MS, MALDI, GC-MS, HPLC, GC, DSC, TGA, XRD, SC-XRD, AFM/STM, NMR, EPR, Microwave Synthesis	44 instruments (see SIL Inventory sheet)	Chem Bldg 016	24/7 badge access (James)
BIOLOGICAL & BIOMEDICAL					
Nevada Genomics Center	School of Medicine / NCAR	Next-gen sequencing, Sanger sequencing, fragment analysis	Illumina NextSeq 2000, MiniSeq, SeqStudio, Agilent TapeStation 4150, QuantStudio 3	School of Medicine	Fee-for-service
Mick Hitchcock Proteomics Center	School of Medicine / NCAR	LC-MS/MS proteomics, gel electrophoresis	Thermo Orbitrap Eclipse + nanoLC, Bruker timsTOF Pro2 + NanoElute, Evosep One	School of Medicine	Fee-for-service
Cell Analysis Core (CACF)	School of Medicine	Flow cytometry, cell sorting	BD Biosciences flow cytometers/sorters	School of Medicine	Training required
Cell Metabolism Core	School of Medicine	Cellular respiration, metabolic analysis	Seahorse XF24, Seahorse XFp analyzers	School of Medicine	Fee-for-service
HSTRI Imaging Core	School of Medicine	High-resolution biological imaging	High-speed imaging systems	School of Medicine	Fee-for-service
Biosciences Entrepreneurial Lab	NCAR	Biotech operations, fermentation, cell culture	BSL cabinets, autoclaves, incubators, freezers	NCAR	Lease-based
Nevada Bioinformatics Center	School of Medicine / NCAR	Bioinformatics, statistical analysis, genomic assembly	Computational infrastructure	School of Medicine	Collaborative
IMAGING & ELECTRON MICROSCOPY					
Cellular & Molecular Imaging (CMI)	Neuroscience / NCAR	Confocal, two-photon, brightfield/fluorescent, 3D tissue imaging	Leica SP8 LIGHTNING confocal, Scientifica two-photon, Leica THUNDER	Neuroscience	Training required
Electron Microscopy & Microanalysis (EMMF)	Engineering / NCAR	S/TEM, EDS, EBSD, FIB-SEM	FEI Talos F200s S/TEM, Thermo Scios 2 FIB/SEM	Engineering	Training + fee
Mackay Microbeam Laboratory	Geological Sciences / NCAR	EPMA, SEM, WDS/EDS, cathodoluminescence	JEOL JXA iHP200F FE-EPMA (5 WDS), JEOL JSM 7100FT, JSM 6010LA	Mackay School	Fee-for-service
Materials Characterization NV (MCNV)	Materials Science / NCAR	TEM, EDS, failure analysis, corrosion testing	JEOL JEM 2100F TEM, Hitachi S-4700 II FE-SEM	Materials Sci.	Fee-for-service
PHYSICS & HIGH-ENERGY					
Zebra Pulsed Power Lab (ZPPL)	Physics	Pulsed power, Z-pinch, high-power lasers, plasma physics	2 TW Z-pinch (Zebra), 50 TW Leopard laser, 100 TW fs laser, 800-core HPC	Physics	Collaborative
MacLean Observatory	Physics	Optical astronomy, CCD imaging	22" and 14" telescopes, CCD cameras, spectrometers	Campus	Physics Dept.
ENGINEERING & MANUFACTURING					
Davidson Foundation Cleanroom	Engineering	Nanofabrication, lithography, deposition, etching	Class 1000/100 cleanroom, litho/deposition/etch/metrology tools	Engineering	Training required
ME Manufacturing Lab	Mechanical Engineering / NCAR	CNC, welding, machining, friction stir welding	Tormach CNC, lathes, mills, FSW machine, hardness tester	ME Bldg	Training required
Environmental Engineering Lab	Civil Engineering / NCAR	GC-MS, LC, environmental analysis	Waters Quattro micro GC-MS, Agilent 1260 LC	Civil Eng.	Fee-for-service
CIVIL / EARTHQUAKE ENGINEERING					
CCEER Earthquake Lab	Civil Engineering	Shake table testing, seismic simulation	4 biaxial + 1 6-DOF shake tables, 80 ft strong wall	CCEER	Collaborative
Western Regional Superpave Center	Civil Engineering / NCAR	Asphalt testing, rheology, pavement performance	2 DSRs, BBRs, AMPT, fatigue machines, FTIR	Civil Eng.	Fee-for-service
ENVIRONMENTAL & FIELD SCIENCES					
Nevada Stable Isotope Lab	Geology / NCAR	Stable isotope analysis (C, O, H, N, S)	Picarro L2130-i, 2x Micromass IsoPrime IRMS, Eurovector EA3000	Geology	Fee-for-service
Valley Road Field Lab	Natural Resources	Greenhouse research, tissue culture	29,280 sq ft greenhouse complex	Valley Road	NRES Dept.
NEUROSCIENCE / MEDICAL IMAGING					
Neuroimaging Core	Neuroscience	fMRI, structural MRI, HD-EEG, fNIRS	2x 3T MRI (Philips Ingenia, Siemens Skyra) at Renown	Renown Health	Collaborative
BSL-3 Lab	School of Medicine	Infectious agent research	BSL-3 containment equipment	School of Medicine	Faculty approval

University of British Columbia — Core Science & Engineering Facilities

Facility Name	Department / Affiliation	Key Technology Categories	Notable Equipment	Notes	Access Notes
CHEMISTRY & SPECTROSCOPY					
Chemistry Shared Instrument Facility (SIF)	Chemistry	NMR (300 MHz), FTIR, UV-Vis-NIR, GC-MS, AFM, TGA, Polarimetry	Bruker Avance-III 300 MHz, PerkinElmer Frontier FTIR, Agilent 6890N GC/5975B MS, Nanosurf AFM, Netzsch TGA209, Jasco P-2000	Open access	Open access for UBC researchers; external users may apply through PI collaboration. Standard instrument training required.
Mass Spectrometry / Microanalysis	Chemistry	High-resolution MS, elemental microanalysis	Multiple MS instruments	Service-based	Service-based (submit samples); open to external academic and industry clients for a fee.
X-Ray Crystallography Facility	Chemistry	Single-crystal XRD, powder XRD, structural databases	SC-XRD and PXRD instruments	Rooms E214-E217	Primarily Chemistry dept. users; external via PI sponsorship. Training for hands-on XRD.
Biological Services Lab	Chemistry	BSL-2 cell culture, mammalian cell assays	Class II Type B2 biosafety cabinets		Restricted to trained UBC researchers; BSL-2 certification required.
ELECTRON MICROSCOPY & IMAGING					
Bioimaging Facility (BIF)	Botany	SEM, TEM, Cryo-SEM/TEM, FIB-SEM, confocal, CLEM	Multiple SEM/TEM instruments, cryo-fixation equipment	BioSci Building	User fee schedule; external academic and industry users accepted. Hourly rates $40-150/hr depending on instrument.
Electron Microscopy Lab (EML)	Materials Engineering	SEM, TEM, EDS, XRD, sample prep	Hitachi SU8700, FEI Quanta 650, Thermo Apreo, Tescan MIRA, FEI Tecnai 120/200 kV	Frank Forward Bldg	External users accepted; fee-for-service or collaboration. Rates differ for academic vs industry.
HR Cryo-EM Facility (HRMEM)	Biochemistry	Cryo-TEM, cryo-FIB-SEM, single particle, tomography	Titan Krios (energy filter), Glacios, Aquilos 2 cryo-FIB (only in Canada), 2x Vitrobot Mark IV		Competitive application; external users accepted via collaboration with UBC PI. Very high demand — typical wait 2-4 weeks.
LIFE SCIENCES					
LSI Imaging Facility	Life Sciences Institute	Widefield, super-resolution (STED/SMLM/SIM), confocal, spinning disk, HT imaging	Multiple advanced light microscopes	LSC Building	UBC researchers with training; external via collaboration. Booking system for instrument time.
ubcFLOW	Life Sciences Institute	Flow cytometry analysis and sorting	6 high-end cell sorters, 7 analysers		Fee-for-service; external academic users accepted. Sorting sessions bookable online.
ubcCyTOF	Life Sciences Institute	Mass cytometry	CyTOF instrument		Fee-for-service; external academic users accepted via LSI.
Sequencing & Bioinformatics Consortium	Multi-department	NGS (Illumina), Sanger sequencing, library prep, bioinformatics	Illumina sequencers		Service-based; accepts external academic samples. Turnaround 2-6 weeks depending on project.
Proteomics Core Facility	Life Sciences Institute	Quantitative proteomics, phosphoproteomics	MS instruments		Service-based; external academic collaboration accepted. Consultation required before submission.
NANOFABRICATION & MATERIALS					
AMPEL Nanofabrication & Cleanroom	Multidisciplinary	ISO-5/6 cleanrooms, photolithography, soft lithography	Heidelberg MLA150, Neutronix-Quintel mask aligner	Frank Forward/Brimacombe	External academic users accepted with training; cleanroom fees ~$50-100/hr. Industry rates higher.
AMPEL High Head Laboratory	Multidisciplinary	Pilot-scale: metal casting, ceramic prep, plasma spray, fluidization	1,500 m^2 pilot lab		Primarily UBC engineering; external via research collaboration.
Advanced Nanofabrication Facility (ANF)	Quantum Matter Institute	Quantum device fabrication, ISO-5/6 cleanrooms	Contact & maskless lithography	QMI Building	Highly restricted; primarily QMI researchers. External access via QMI affiliation or collaboration.
PHYSICS & QUANTUM					
Physics & Astronomy Labs	Physics	MBE, UHV-STM, XRD, cryogenics, cleanrooms	3He/4He refrigerators, MBE system, UHV-STM, ~1000 node HPC	Hennings/Hebb	Primarily Physics dept.; some shared access for collaborators.
Quantum Matter Institute (QMI)	Multidisciplinary	Vibration-free microscopy, quantum materials characterization	Underground vaults, environmental control, atomic-scale spectroscopy	QMI Building	Restricted to QMI members and approved collaborators.
AQUATIC & ENVIRONMENTAL					
Aquatic Ecosystems Research Lab (AERL)	Fisheries/IRES	Wet labs, dry labs, ecosystem modeling	53,000 sq ft facility		UBC researchers and approved external collaborators.
InSEAS	Environmental	Aquaculture labs, environmental chambers	14 aquatic labs, 10 controlled chambers		UBC researchers; some external academic access.
Wine Research Centre MS Core	Food/Nutrition	Organic mass spectrometry	9 MS instruments, multiple ionization modes		Service-based; accepts external academic and industry samples.
OTHER					
Centre for Comparative Medicine	Veterinary	Large animal vivarium (largest in Canada)	Multi-species: primates, pigs, rabbits, reptiles	South campus	Restricted; CCAC-approved protocols required. External via UBC PI collaboration.
Clean Energy Research Centre	Chem/Bio Engineering	Bioenergy, battery research, district energy	BRDF demonstration facility	ChBE Building	Primarily internal; external via research collaboration agreements.
Centre for Blood Research	Chemistry/Medicine	Automation/HTS, X-ray crystallography, biophysics	Biofactorial facility, crystallography hub	LSC 4th floor	Internal researchers; external via collaboration with CBR faculty.

Massachusetts Institute of Technology — Core Science & Engineering Facilities

Facility Name	Department / Affiliation	Key Technology Categories	Notable Equipment	Notes	Access Notes
NANOFABRICATION & CHARACTERIZATION					
MIT.nano (Fab + Characterization)	Institute-wide	Nanofab: litho, etch, deposition; Characterization: AFM, XRD, SEM, TEM, cryo-EM, surface analysis	160+ tools; Titan Krios, Talos Arctica, Aquilos cryo-FIB; e-beam litho (20 nm)	50+ core facilities total	Open to all MIT affiliates; external academic and industry users accepted ($250-500/hr+). Online reservation system.
Microsystems Technology Labs (MTL)	EECS/Interdept.	Class-10 cleanroom, photolithography, MEMS, microelectronics fab	6,500 sq ft cleanroom; equipment value >$50M		MIT community; external via sponsored projects. Cleanroom training required (multi-day).
SPECTROSCOPY & DIFFRACTION					
Dept. Chemistry Instrumentation (DCIF)	Chemistry	NMR (7 instruments, 400-600 MHz, 4 w/ cryoprobes), MS, EPR	Bruker NMR spectrometers, Bruker EMX-Plus EPR (4.5K cryo)	24/7 access; Bldg 18	24/7 walk-up access for trained MIT researchers; external academic users accepted with PI sponsorship. Walk-up NMR available.
Francis Bitter Magnet Laboratory	PSFC	Solution/solid-state NMR, EPR, Dynamic Nuclear Polarization	High-field magnets, DNP instrumentation		MIT researchers; external via collaboration. Specialized training for DNP.
X-Ray Diffraction Facility	Chemistry	Single-crystal XRD (small molecule + protein)	Bruker APEX/APEX2 (Cu + Mo); 90-500K temp control; 10 µm crystals	Bldg 2, Room 325	MIT Chemistry; external academic users accepted. Submit crystals for service or train for self-service.
ELECTRON MICROSCOPY					
Cryo-EM Facility	MIT.nano / Biology	Cryo-TEM, cryo-SEM, cryo-FIB	Titan Krios, Talos Arctica, Aquilos cryo-FIB/SEM, Vitrobot		MIT affiliates; external academic users via MIT.nano membership. High demand — book well in advance.
Electron Microscopy Characterization Lab	Materials Science	SEM, magnetic/electronic property measurement	SEM instruments, property measurement tools	DMSE	DMSE community; external via collaboration.
GENOMICS & SEQUENCING					
BioMicro Center	Biology / Koch Institute	Illumina short-read, PacBio/Nanopore long-read, single-cell, qPCR, microarray	NextSeq, HiSeq, MiSeq, PacBio, Nanopore, liquid handlers	Bldg 68	Fee-for-service; open to MIT and external academic researchers. Full-service or assisted options.
Broad Institute Genomics Platform	MIT/Harvard	Massive-scale sequencing (12 TB/day; 1 genome per 12 min)	Illumina NovaSeq fleet; 100,000 sq ft facility	320 Charles St	Open to MIT/Harvard affiliates and external academic collaborators; fee-for-service. Very high throughput.
FLOW CYTOMETRY & CELL ANALYSIS					
Koch Institute Flow Cytometry Core	Koch Institute	High-speed sorting, nanowell cytometry, microengraving, SeqWell	BD, Sony sorters; nanowell platforms		Koch Institute affiliates; external via Koch PI collaboration.
Whitehead Institute Flow Cytometry	Whitehead / MIT	Flow analysis and sorting	Multiple cytometers; 24/7 automated access		Whitehead members; external via collaboration. 24/7 automated access for trained users.
PROTEOMICS & METABOLOMICS					
Koch Biopolymers & Proteomics Core	Koch Institute	Protein synthesis, small molecule analysis, nanoparticle characterization	MS instruments		Koch Institute affiliates; external via Koch PI collaboration.
Whitehead Quantitative Proteomics	Whitehead / MIT	Mass spectrometry proteomics	High-resolution MS instruments		Whitehead members; external via collaboration.
CEHS Chromatography & MS Core	Env. Health Sciences	LC-MS/MS, chemical separation	LC-MS/MS instruments		MIT community; external via collaboration.
Whitehead Metabolite Profiling	Whitehead / MIT	Metabolomics, small molecule MS	MS instruments		Whitehead members; external via collaboration.
OPTICAL MICROSCOPY & IMAGING					
W.M. Keck Imaging Facility	Whitehead / MIT	Confocal, two-photon, live-cell imaging	Advanced microscopy suite		Whitehead members; external via collaboration. Training required.
Koch Institute Microscopy Core	Koch Institute	Advanced light microscopy, image analysis	Confocal and widefield instruments		Koch Institute affiliates; external via Koch PI collaboration.
Martinos Imaging Center	McGovern Institute	Human 3T MRI, 9.4T animal MRI, fMRI, EEG, MEG, fNIRS	3T Siemens, 9.4T Bruker, MEG system	McGovern	MIT and affiliate researchers; external via collaboration. IRB approval required for human imaging.
NUCLEAR & PLASMA					
MIT Nuclear Reactor Lab (MITR)	Nuclear Science	6 MW research reactor, neutron irradiation, hot cells, radiochemistry	6 MW reactor; beam ports; SEM, optical microscopy in-house	2nd largest US university reactor	Restricted facility; requires NRC licensing. Access through MIT Nuclear Science. Unique resource.
Plasma Science & Fusion Center	PSFC	Tokamak, pulsed power (PUFFIN), superconducting magnets, fusion materials	Alcator C-Mod (legacy), cyclotron, Schmidt Lab		MIT researchers; some external collaborations. Highly specialized.
MATERIALS & OTHER					
Materials Research Lab (MRL)	Interdepartmental	Materials characterization, X-ray, thermal analysis	Shared characterization instruments		MIT community; shared instrument access with training.
DMSE Core Facilities	Materials Science	Prototyping (LEM), metallography, SEM, property measurement	Machine shop, metallurgical prep, SEM		DMSE community; machine shop and characterization open to trained users.
Lincoln Lab Microelectronics	Lincoln Laboratory	200 mm wafer fab: CMOS, CCD, superconducting, photonics, MEMS	70,000 sq ft fab; Class-10/100 cleanrooms; 24/5 operations	Hanscom AFB	DOD-funded; restricted access. Primarily Lincoln Lab employees and sponsored researchers.
Lincoln Lab Supercomputing	Lincoln Laboratory	HPC, machine learning, AI	New England's most powerful supercomputer	Holyoke, MA	Restricted; Lincoln Lab and approved collaborators only.

California Institute of Technology — Core Science & Engineering Facilities

Facility Name	Department / Affiliation	Key Technology Categories	Notable Equipment	Notes	Access Notes
BECKMAN INSTITUTE FACILITIES					
Biological Imaging Facility (BIF)	Beckman Institute	Confocal, light sheet, widefield fluorescence, FLIM, deconvolution	5 Zeiss LSMs, LSM 980 Airyscan 2, Leica STELLARIS 8 FALCON/DIVE, 2 LaVision light sheet, Nikon spinning disk	Room B133	Open to all Caltech researchers with training; external via PI collaboration. Online booking.
X-Ray Crystallography Facility	Beckman Institute	Single-crystal XRD, powder XRD (XRPD)	2 Bruker AXS SC-XRD (D8 KAPPA, D8 VENTURE), Rigaku SmartLab, PANalytical X'Pert Pro	Room BI110	Caltech researchers with training; external academic users via collaboration. Self-service after training.
Cryo-EM Center	Beckman Institute	Cryo-TEM, cryo-SEM-FIB, structural biology	FEI Titan Krios 300 kV + Gatan K3 detector + BioQuantum filter; FEI Versa cryo-SEM-FIB		Caltech researchers; external via collaboration. Competitive scheduling — book 1-2 weeks ahead.
Flow Cytometry / Cell Sorting	Beckman Institute	Flow cytometry, high-speed cell sorting	Sony SY3200, BD FACS Aria Fusion, Beckman Cytoflex SRT, BioSorter, MACSQuant VYB	30+ labs served	Caltech community; external via collaboration. Serves 30+ labs — high demand.
Molecular Observatory	Beckman Institute	Macromolecular crystallography, protein crystallization	Rigaku MicroMax 007-HF, RAXIS-IV++ detector, Oxford Cryostream; SSRL beamline access		Primarily Caltech structural biology groups; external via collaboration. SSRL beamline access available.
Proteome Exploration Lab (PEL)	Beckman Institute	LC-MS/MS proteomics, protein modifications, protein-protein interactions	3 Thermo Orbitraps: QExactive HF, Fusion Tribrid, Eclipse Tribrid (UVPD/FAIMS)		Fee-for-service and collaborative; accepts external academic samples. Consultation before submission.
Single-Cell Profiling Center (SPEC)	Beckman Institute	Single-cell genomics, scRNA-seq	Illumina MiSeq on-site; NovaSeq outsourced		Caltech researchers; external via collaboration. Full-service or assisted sequencing.
Zebrafish/Xenopus/Lamprey Facility	Beckman Institute	Model organism housing, transgenic generation, microinjection	1000 zebrafish tanks, 4 lamprey tanks, microinjectors, fluorescence microscopes		Caltech researchers; BSL-1/2 trained personnel only.
CHEMISTRY & CHEMICAL ENGINEERING					
Liquid NMR Facility	CCE	1H, 13C, 19F, 15N, 31P NMR	2x 300 MHz, 1x 300 MHz autosampled, 1x 400 MHz, 2x 500 MHz, 1x 600 MHz penta probe	Crellin B250	Walk-up and scheduled access for trained Caltech researchers. 7 instruments, generally good availability.
Solid-State NMR Facility (CSNF)	CCE	Solid-state NMR spectroscopy	Bruker Avance 200, AM 300, Avance 500		Caltech researchers with training; specialized solid-state NMR expertise required.
Multi-User Mass Spec Lab	CCE	GC-MS, MALDI, EI/CI ionization, accurate mass	JEOL AccuTOF GC-Alpha (high-res GC-MS), MALDI instrument	Crellin B170	Self-service for trained Caltech users; external via collaboration. Open access GC-MS.
Center for Catalysis & Chemical Synthesis	CCE	Catalyst development, chemical synthesis	Synthesis infrastructure		Caltech CCE researchers; generally open to collaborators.
KAVLI NANOSCIENCE INSTITUTE (KNI)					
KNI Laboratory	KNI	E-beam litho, ion beam litho, optical litho, two-photon litho, nanofab, characterization	2x 100 kV e-beam writers, Heidelberg optical, Sirion FE-SEM (<10 nm), FEI Versa cryo-SEM-FIB; 10,000 sq ft cleanroom	Open to external users	Open to external academic and industry users (rare for a university cleanroom). Fee schedule available. Training required.
GEOLOGICAL & PLANETARY SCIENCES					
GPS Division Analytical Facility	GPS	SEM, EPMA, XRF, mass spec (plasma/gas/SIMS), Raman, IR, high-P/T apparatus	PANalytical Zetium XRF (6 crystals); electron microprobe; diamond anvil, multi-anvil, shock-wave		GPS division researchers; external academic users via collaboration.
Center for Microanalysis	GPS	SIMS, NanoSIMS, ion imaging, depth profiling, geochronology	Cameca NanoSIMS 50L (50 nm, 7 masses), Cameca IMS 7f-Geo	Unique: sub-50 nm isotopic imaging	Caltech GPS; external via collaboration. NanoSIMS is a unique capability — few instruments worldwide.
Resnick Water & Environment Lab	Linde Center	GC/LC/IC-MS, ICP-MS, TOC/TIC, isotope ratios, particle sizing	3x ICP-MS, GC-MS, LC-MS, IC-MS, IRMS, particle sizer	User-reservation access	User-reservation access; open to Caltech researchers. External via collaboration.
Linde Center Atmospheric Chambers	GPS/Env. Science	Atmospheric chemistry, photochemistry, aerosol characterization	2x 1,000 ft^3 reaction chambers, high-precision laser spectroscopy	Acoustic/EM insulated	Caltech researchers; specialized atmospheric science community.
MATERIALS SCIENCE & TEM					
TEM Facility	Materials Science	High-resolution TEM, EDS, HAADF, sample prep	FEI Tecnai F30ST 300 kV, Oxford EDS, Gatan UltraScan camera	Keck Eng. Lab B042	Caltech materials science; external via collaboration. Training required for hands-on.
Transgenic Mouse Facility	Biology	Mouse genetic engineering and breeding	Transgenic generation equipment		Restricted; Caltech biology with IACUC approval required.
LARGE-SCALE OBSERVATORIES					
LIGO Laboratory	Caltech/MIT	Gravitational wave detection via laser interferometry	2 detectors: LIGO Hanford (WA), LIGO Livingston (LA)	80+ partner institutions	Caltech/MIT joint operation; open to broader scientific community through proposals.

UNR SIL Coverage vs. Total Technology Universe

Estimated coverage of your current 24/7 SIL access against the combined technology landscape

Technology Domain	Total Sub-Categories (Universe)	Available at UNR SIL	SIL Coverage %	Gap / What You're Missing
Vibrational Spectroscopy	4	4	100%	Full coverage: FT-IR, micro-FTIR, Raman, ATR
Electronic Spectroscopy	5	5	100%	Full coverage: UV-Vis, UV-Vis-NIR, fluorescence, CD, polarimetry
Mass Spectrometry	9	4	44%	Missing: LC-MS/MS (Orbitrap), ICP-MS, SIMS/NanoSIMS, IRMS, ion chromatography-MS
Chromatography / Separation	3	2	67%	Missing: Ion Chromatography
NMR Spectroscopy	5	4	80%	Missing: Dynamic Nuclear Polarization (DNP)
EPR Spectroscopy	2	2	100%	Full coverage: X-band EPR + variable-temperature
Thermal Analysis	5	3	60%	Missing: DMA, Rheometry
X-Ray Techniques	5	2	40%	Missing: XRF, synchrotron access, macromolecular crystallography
Electron Microscopy	6	0	0%	No SEM, TEM, cryo-EM, FIB-SEM, EPMA — available at UNR EMMF/Mackay
Surface Analysis / Microscopy	4	2	50%	Missing: XPS/ESCA, profilometry
Optical / Fluorescence Microscopy	6	0	0%	No confocal, two-photon, light sheet, super-resolution — UNR CMI has some
Genomics & Sequencing	5	0	0%	No NGS, long-read, Sanger, single-cell, qPCR — UNR Genomics Center
Proteomics & Metabolomics	3	0	0%	No quantitative proteomics, metabolomics, protein crystallography — UNR Proteomics Ctr
Flow Cytometry & Cell Analysis	4	0	0%	No flow cytometry, FACS, CyTOF, Seahorse — UNR CACF/Cell Metab Core
Nanofabrication & Cleanroom	6	1	17%	SIL has Denton evaporator only; missing litho, e-beam, etch, MBE, microfluidics — UNR Davidson Cleanroom
Synthesis & Reaction Support	4	3	75%	Missing: High-pressure apparatus (diamond anvil, etc.)
Materials Testing & Fabrication	6	3	50%	Missing: mechanical testing, CNC machining, FSW — UNR ME Lab
Nuclear & Radiation Research	4	0	0%	No reactor, neutron sources; UNR ZPPL has pulsed power/lasers
Brain & Medical Imaging	4	0	0%	No MRI, EEG, MEG — UNR Neuroimaging Core at Renown
Astronomy & Observatories	3	0	0%	UNR MacLean Observatory nearby; LIGO is MIT/Caltech
Environmental & Geoscience	4	0	0%	No isotope analysis, atmospheric chambers, seismic — UNR has separate labs
Sample Preparation & Infrastructure	4	4	100%	Full coverage: ultrapure water, N_2/He gas, balances, UPS
Bioinformatics & Computing	2	0	0%	UNR Bioinformatics Center available separately
Biological Model Systems	4	0	0%	No animal facilities, BSL-3 — UNR School of Medicine

OVERALL SUMMARY

Total technology sub-categories across all institutions:	**105**
Sub-categories available at UNR SIL:	**39**
UNR SIL COVERAGE OF TOTAL UNIVERSE:	**~37%**
UNR SIL + Other UNR Core Facilities combined:	**~65-70%**
STRONGEST COVERAGE (SIL alone):	**Vibrational spectroscopy, electronic spectroscopy, EPR, NMR, sample prep**
BIGGEST GAPS (SIL alone):	**Electron microscopy, genomics, proteomics, flow cytometry, brain imaging, cleanroom**
BIGGEST GAPS (even across all UNR):	**Cryo-EM, NanoSIMS, DNP-NMR, synchrotron, nuclear reactor, long-read sequencing**